

Vulnerability Assessment Report

18th May 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Consider the following questions to help you write:

The purpose of this vulnerability assessment is to identify weaknesses regarding the critical factors that could cause direct harm to business processes in regards to its technical infrastructure. The predominant asset that is in danger is the database server of the business, it holds all critical information pertaining to the business and holds a single point of failure making its security utmost importance. In the chance that the server was corrupted, exfiltrated or destroyed, business continuity could be impacted and trust from customers would falter.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	Perform reconnaissance and surveillance	2	2	4
Hacker	Alter/ Delete Critical information	1	3	3
Humidity controls/temper	Disrupt Mission critical operations	2	3	6

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Approach

This vulnerability report was written around the context of memory being the primary target and point of failure in the case of a security incident. In business contexts, it is quite a likely outcome that competitors will try and obtain insider information of a business, given that they obtain the information it is unlikely they will compromise business processes but rather use that information to benefit them without direct hindrances to avoid malpractice suits. The single point of failure of the database server, poses a threat from hackers as there is no cloud or outsourced storage solutions, if data is altered or deleted it could pose serious issues to the business though the likelihood is low due to the business being small and very low return on investment for a hacker. Finally the most likely vulnerability comes from natural causes such as humidity or temperature affecting data which due to single point of failure design, has high repercussions and is a far more likely occurrence than any attack or reconnaissance from a competitor or hacker.

Remediation Strategy

The remediation process is interconnected due to one single point of failure being the cause of many vulnerabilities. To limit competitor reconnaissance, the principle of least privilege can be applied so that the amount of employees that have access to confidential data and the attack vector of competitors finding out through insider leakage is very low. To combat hackers and the chances of alterations to data, high level encryption can be used. Any outdated or simple hashing and encryption algorithms should be replaced as this is a cost effective and effective method of protecting critical data points. For physical threats like environmental factors, regular system checkups like physically checking the health of storage devices from any signs of humidity or dust or physical damage of any sort can be a good prevention of data damage. The most effective form of data protection in all three vulnerabilities however would be to invest in a cloud storage solution. This would add third party vendor security of the data stopping hackers and environmental factors while still applying principle of least privilege.